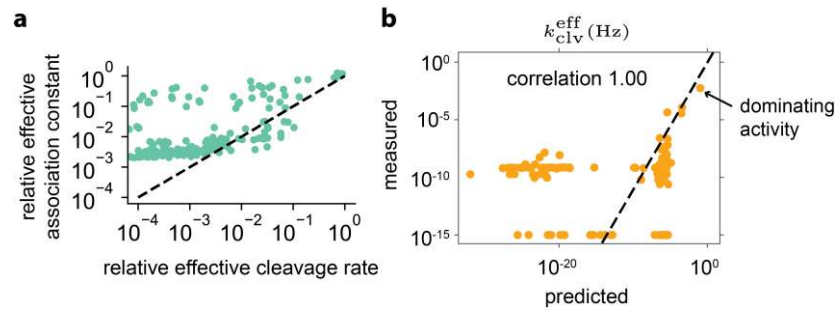
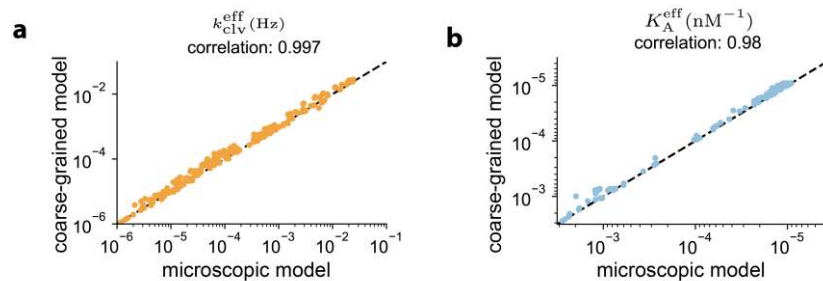


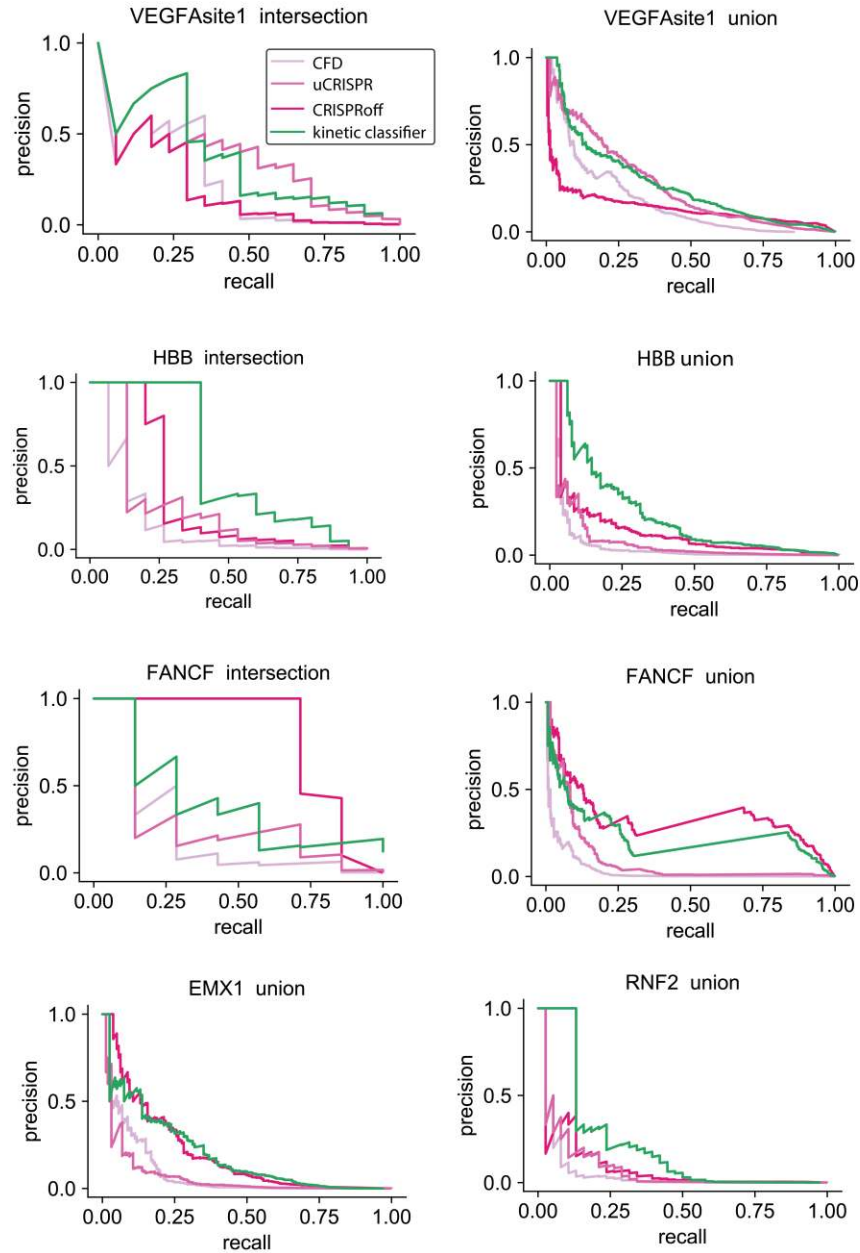
Supplementary Information for
A kinetic model predicts *SpCas9* activity, improves off-target classification, and reveals the physical basis of targeting fidelity



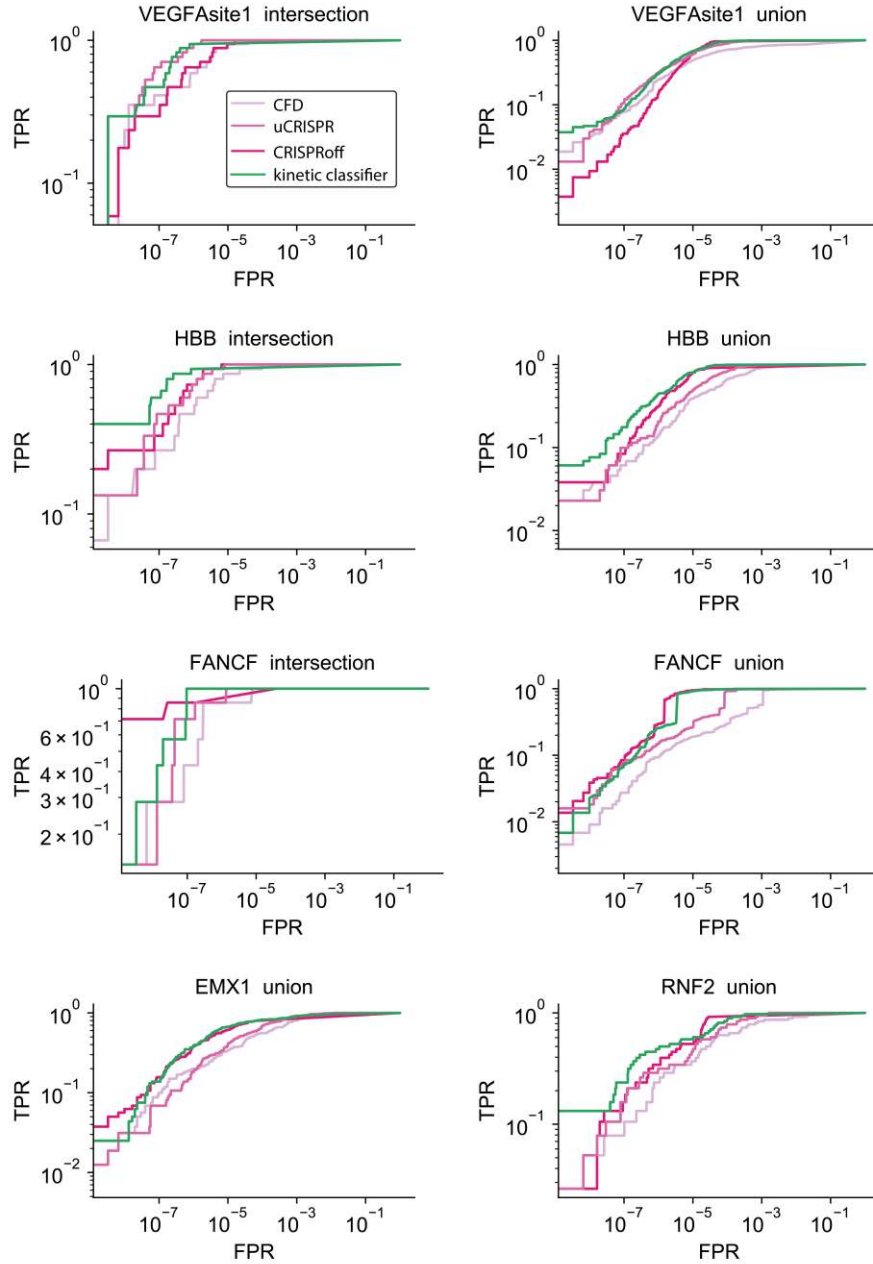
Supplementary Figure 1 **a)** Correlation plot of measured effective cleavage rates (NucleaSeq) relative to on-target, versus effective association constants (CHAMP) relative to on-target. The dashed line indicates where these relative measures would be equal. **b)** Correlation plot of effective cleavage rate measurements (NucleaSeq) versus model predictions for highly mismatched off-targets (three or more mismatches). The model successfully separates out the single target that dominates the Pearson-correlation calculation, resulting in a perfect Pearson correlation. The dashed line represent perfect quantitative prediction.



Supplementary Figure 2 **a)** Correlation between predictions for effective cleavage rate using the microscopic kinetic model (**Figure 1a**) and the coarse-grained kinetic model (**Figure 5a**). **b)** Same as **a**, for effective association constants. The very high correlation for both quantities shown that the coarse grained model retains the predictive power of the microscopic model. The dashed line represent perfect quantitative correspondence between models.



Supplementary Figure 3 a) Precision-recall curves for the CFD score (light purple), uCRISPR score (purple), CRISPRoff (dark purple), and our kinetic classifier (green) for the FANCF, VEGFA site 1, EMX1, HBB and RNF2 target sites. We use union to denote curves where all experimentally identified off-targets are used as true positives, and intersection to denote curves where off-targets identified in all experiments are used as true positives.



Supplementary Figure 4. Receiver-operator characteristic curves for the CFD score (light purple), uCRISPR score (purple), CRISPRoff (dark purple), and our kinetic classifier (green) for the FANCF, VEGFA site 1, EMX1, HBB and RNF2 target sites. We use union to denote curves where all experimentally identified off-targets are used as true positives, and intersection to denote curves where off-targets identified in all experiments are used as true positives. Our data sets are unbalanced, with many more true negatives than true positives. This results in a true-positive rate (TPR) that increases very rapidly with the false-positive rate (FPR) (note the log-scale used). This renders the area under the curve very close to one in every case, and to allow us to differentiate between models we instead choose to base our analysis in precision-recall curve (Supplementary Figure 3).

target	Technique and reference	guide sequence
EMX1	BLESS ¹	GAGTCCGAGCAGAAGAAGAAGGG
	BLISS ²	GAGTCCGAGCAGAAGAAGAAGGG
	CIRCLE-seq ³	GAGTCCGAGCAGAAGAAGAANGG
	Digenome-seq ^{4,5}	GAGTCCGAGCAGAAGAAGAAGGG
	GUIDE-seq ⁶	GAGTCCGAGCAGAAGAAGAANGG
	HTGTS ⁷	GAGTCCGAGCAGAAGAAGAAGGG
FANCF	CIRCLE-seq ³	GGAATCCCTTCTGCAGCACCNNGG
	Digenome-seq ^{4,5}	GGAATCCCTTCTGCAGCACCTGG
	GUIDE-seq ⁶	GGAATCCCTTCTGCAGCACCNNGG
	SITE-seq ⁸	GGAATCCCTTCTGCAGCACCTGG
HBB	CIRCLE-seq ³	GTTGCCCCACAGGGCAGTAANGG
	Digenome-seq ^{4,5}	CTTGCCCCACAGGGCAGTAACGG
RNF2	CIRCLE-seq ³	GTCATCTTAGTCATTACCTGNGG
	Digenome-seq ^{4,5}	GTCATCTTAGTCATTACCTGAGG
	GUIDE-seq ⁶	GTCATCTTAGTCATTACCTGNGG
VEGFA_site1	CIRCLE-seq ³	GGGTGGGGGGGAGTTTGCTCCNNGG
	Digenome-seq ^{4,5}	GGGTGGGGGGGAGTTTGCTCCAGG
	GUIDE-seq ⁶	GGGTGGGGGGGAGTTTGCTCCNNGG
	HTGTS ⁷	GGGTGGGGGGGAGTTTGCTCCTGG
	SITE-seq ⁸	GGGTGGGGGGGAGTTTGCTCCTGG

Supplementary Table 1: Data sets used to collect true positives for whole genome analysis.

Supplementary references

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